



HPC Decoupling – ROF Scope Project

Dry Type Transformer Foundation Design

DOCUMENT NO.
RED-00-ST-CAL-0096

Discipline Seal Applied For Rev 0	Permit to Practice Seal Applied for Rev 0

0	June 29, 2021	Issued for Information – Project 20170	C. Walsh	B. Kapcsos	N/A	N/A
Rev	Date of Issue	Revision Purpose	Prepared By	Checked By	Approved By	IPPL Acceptance

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Location: Sturgeon County, Alberta
Title: Dry Type Transformer Foundation Design
Calculation No.: RED-00-ST-CAL-0096
By: Coltin Walsh

Contact No.: A6KV
Date: 29-Jun-2021
Check Level: 2B
Chk'd: Brett Kapcsos

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1.0 DESIGN SUMMARY

1.1 Purpose of Calculation

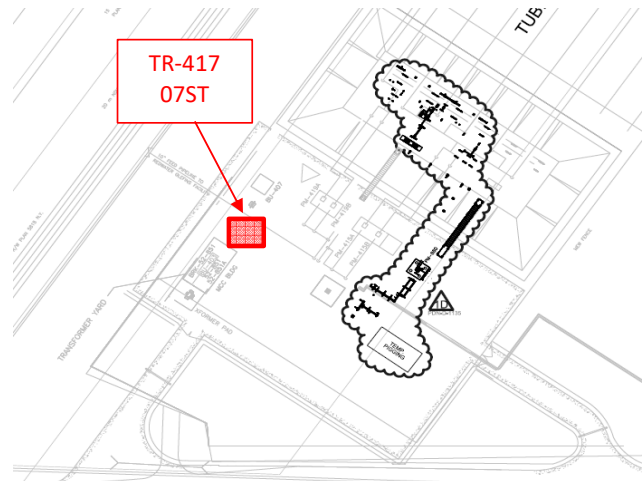
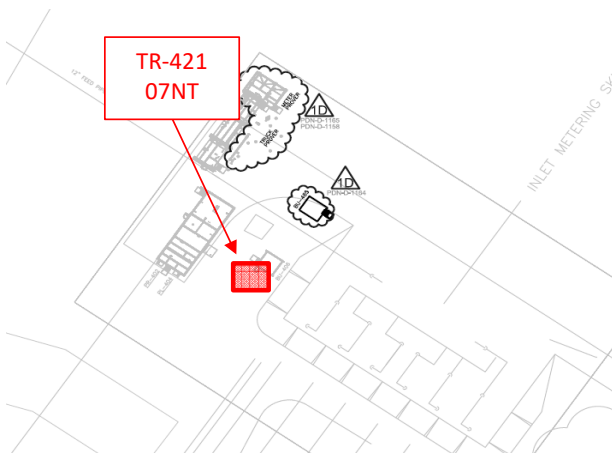
The purpose of this calculation is to design the foundation and support steel for two dry type transformers: TR-421 in 07NT and TR-417 in 07ST. TR-421 is location on an existing steel transformer support, and TR-417 is supported on a new concrete foundation on grade.

1.2 Scope of Calculation

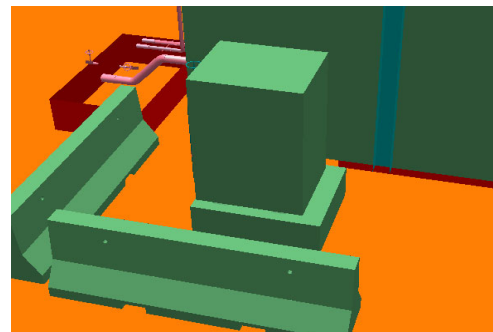
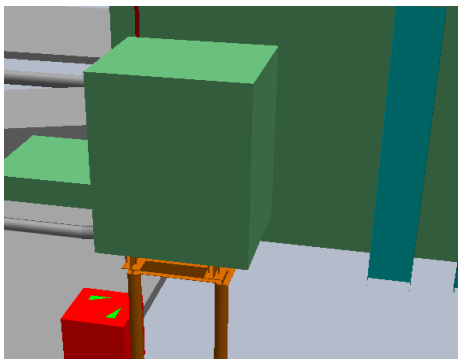
This calculation report includes the following items:

1. Standard load development
2. Structure steel and concrete foundation design

1.3 Structure Layout and Location



S3D Screenshots:





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1.4 Method of Analysis

Structural steel calculations and analysis were performed manually in Excel and using RISA 3D.

1.5 Design Criteria

Design criteria is based on specification PDH-00-ST-SPC-0001 - Structural Engineering Design Criteria.

1.5.1 General

A. Climatic Data

Site meteorological and geological data for the facility is as follows:

As per PDR114_r3, seismic analysis is to be done in accordance with ABC 2014, which references NBCC 2010.

Seismic data extracted from NBCC 2010, Appendix C for Fort Saskatchewan area.

Ground Snow Load, S_s	=	1.6 kPa
Ground Rain Load, S_r	=	0.1 kPa
Hourly Wind Pressure (1-in-50 Year)	=	0.43 kPa
$S_a(0.2)$	=	0.10
$S_a(0.5)$	=	0.06
$S_a(1.0)$	=	0.03
$S_a(2.0)$	=	0.01
PGA	=	0.04
Site Class	=	D

B. Importance Factors

As per Jdb. 2.20, the site will have importance factor: **HIGH**.

Load	ULS	SLS
Wind, I_w	1.15	0.75
Snow, I_s	1.15	0.90
Earthquake, I_e	1.30	-

1.5.2 Governing Codes

The following codes were used for analysis and design:

NBC 2010	National Building Code of Canada 2010
ABC 2014	Alberta Building Code 2014
CSA S16-09	Design of Steel Structures
CSA A23.3-04	Design of Concrete Structures

1.5.3 Specifications and Guidelines

The following specifications and guidelines were used for analysis and design:

Specifications:

Jdb. 2.20 - Piperack Structures - Loads and Combinations
 Propane Dehydrogenation Project specifications and standards
 11073-301.2-SPC-0012 - General Specification for Painting

Geotechnical Report

Redwater Olefin Fractional Plant Geotechnical Investigation - July-31-18



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1.5.4 Strength and Serviceability Requirements

The structure was designed based on industry limits and the requirements specified in the above specifications.

1.5.5 Materials

See standard drawing PDH-SD20-00001 for steel material.
 See standard drawing PDH-SD60-00001 for concrete material.
 See standard drawing PDH-SD60-00012 for helical piling material.

1.6 Software

Microsoft Excel - for calculations and organization of calculation report
 RISA Foundation - for analysis of concrete slab on grade

1.6 Electronic Files

File Name	Description	Directory
RED-00-ST-CAL-0096_R0.pdf	Calculation report	\\fdnet.com\ca_projects\CA_A6KV\215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_014_R_Proj_Calculations_Sketches\RED-00-ST-CAL-0096 - Dry Type Transformer Foundation Design
RED-00-ST-CAL-0092_R0.xls	Calculation report	Same as Calculation Report
TR-417 Foundation.fnd	RISA Foundation File	Same as Calculation Report
TR-417 - Anchorage.pe	Hilti Profis	Same as Calculation Report

2.0 ASSUMPTIONS AND UNCERTAINTIES/RISKS

2.1 Assumptions

ID	Description
N/A	

2.2 Uncertainties and Risks

Uncertainty	Mitigation	Remaining Risk	Comment	Resolution
N/A				

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3.0 Load Development and Foundation Design

3.1 Design Cases

3.1.1 Ultimate Limit States (ULS) Design Cases

A. Operating Condition

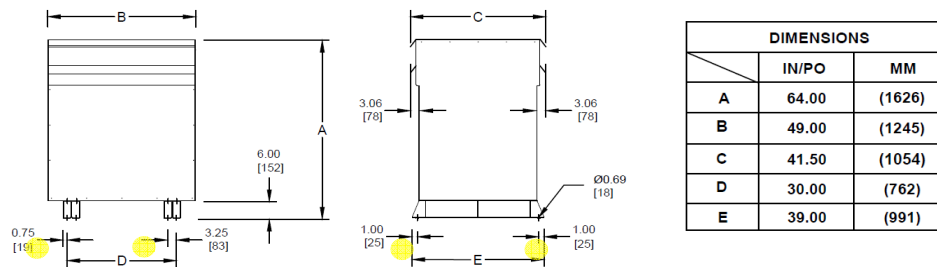
3.1.2 Serviceability Limit States (SLS) Design Cases

A. Settlement Condition
 B. Foundation Overturning

3.2 Basic Load Cases

Both of the dry type transformers (TR-421 in 07NT and TR-417 in 07ST) have identical dimensions and nearly identical weights. Therefore, the load case definition needs to only be completed once below.

The loading from the Transformer is applied at the 4 mounting locations:



Transformer Dimensions:

L = 1245 mm : Length, B
 W = 1054 mm : Width, C
 H = 1626 mm : Height, A
 A_x = 679 mm : Assumed anchor bolt spacing in x-direction, along length (D - 83mm)
 A_z = 941 mm : Assumed anchor bolt spacing in z-direction, along width (E - 50mm)

3.2.1 BLC1: Dead Load, DL

Total dead load includes the weight of the transformer and weight of concrete foundation:

$DL_{\text{transformer}}$ = 10.27 kN : TR-421 = 1047 kg per Vendor Drawing CM6C0225 S023623; use 1047kg.
 : TR-417 = 1006 kg per Vendor Drawing CM6C0225 S023624

DL / Anchor = 2.57 kN

** It is assumed that the main cable tray support will be off of the adjacent building.

Any miscellaneous cable weight added to the transformer foundation is considered negligible and can be absorbed by the current design.

3.2.2 BLC 2: Snow Load, SL

Snow load was calculated using the above climatic data and the NBC 2015:

$S = I_s[S_s(C_b C_w C_s C_a) + S_r]$: NBC 2015
 I_s = 1.15 : Importance factor for ULS
 I_s = 0.90 : Importance factor for SLS
 S_s = 1.60 kPa
 S_r = 0.10 kPa < $S_s(C_b C_w C_s C_a)$ 1.28
 C_b = 0.80 : Basic roof snow load; 0.8 for small roofs with $L_c < 70m$
 C_w = 1.00 : Wind exposure factor, 1.0 for exposed

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$C_s = 1.00$: Slope factor, 1.0 for flat surfaces
 $C_a = 1.00$: Shape factor, 1.0 for roof slope less than 15 degrees
 Blanket Snow Load: $S = 1.38 \text{ kPa}$ > 1 kPa
 $S = 1.59 \text{ kPa}$: $I_s = 1.15$
 $A_{\text{snow}} = 1.3 \text{ m}^2$: Area for snow load application, gross area of the transformer
 $SL = 2.1 \text{ kN}$
 $SL / \text{Anchor} = 0.52 \text{ kN}$

3.2.3 BLC 3: Wind Transverse, WT

Transverse wind is assumed acting in the E/W direction (Z - axis).

$p = I_w q C_e C_g C_p$: Effective wind pressure
 $I_w = 0.43$: Site importance factor, ULS
 $q = 0.43 \text{ kPa}$: 1/50 yr wind pressure
 $C_e = (h/10)^{0.2}$: Exposure factor
 $C_e = 0.90$
 $C_g = 2$
 $C_p = 2$: Conservatively assumed 2.0 for all elements
 $h = 1.63 \text{ m}$
 $w = 0.67 \text{ kPa}$

Transformer Tag	H	L	W	WT [kN]	M _{WT} [kN.m]	WT / Anchor [kN]	T/C* Moment [kN]	*T/C per anchor
Transformer, General	1.626	1.245	-	1.35	1.10	0.34	0.58	

3.2.4 BLC 4: Wind Longitudinal, WL

Longitudinal wind is assumed acting in the N/S direction (X-axis).

WL is applied in the same manner as WT above.

Transformer Tag	H	L	W	WL [kN]	M _{WL} [kN.m]	WL / Anchor [kN]	T/C* Moment [kN]	*T/C per anchor
Transformer, General	1.626	-	1.054	1.14	0.93	0.29	0.68	

3.3 Foundation / Structure Design:

3.3.1 Anchor / Mounting Bolt Check:

The resulting tension and shear acting on the mounting bolts is the same for both transformers:

Critical Tension and Shear from LC: $0.9DL + 1.05WT + 1.05WL$:

$T_f = 0.98 \text{ kN}$
 $V_f = 0.65 \text{ kN}$

As can be seen, the tension and shear at each anchor location is minor.

Each anchor location consists of 2 anchor bolts:

TR-421 = 5/8" ASTM A325 Anchor bolts:

From Table 3-4 in S16-09:

$T_r = 98.0 \text{ kN}$
 $V_r = 54.9 \text{ kN}$

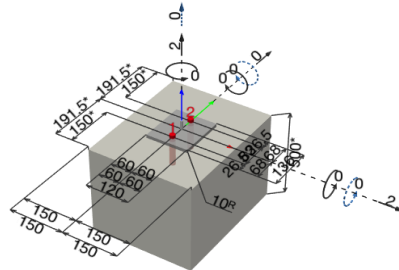
Therefore, the mounting bolts are acceptable by inspection.

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TR-417 = 5/8" C1 Hilti anchors, Embedment = 80 mm:

From Hilti Profis Design:

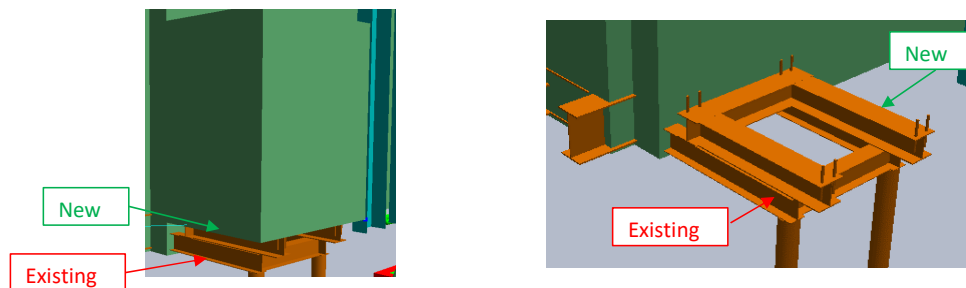


Loads	Forces [kN]			Moments [kNm]			Utilization			
	Vx	Vy	N	Mx	My	Mz	T	S	C	Max
Design	2	0	2	0	0	0	19%	13%	10%	19%

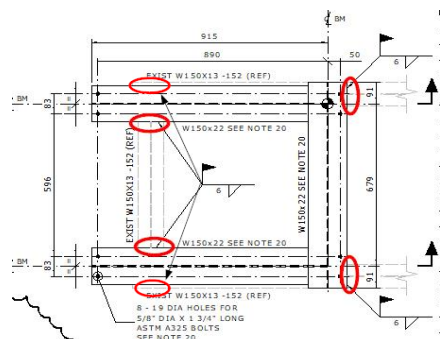
Therefore, the mounting bolts are acceptable by inspection.

3.3.2 TR-421 Steel Mounting Design

Transformer TR-421 is to be mounted on existing steel and pile foundation outside of BU-406 in the North Tubes area. The existing steel was previously used to support a transformer of slight smaller dimensions. Due to the difference in Transformer size and mounting bolt locations, an additional steel frame will be supplied on top of the existing steel:



As per above, the maximum anchor loading is minimal. Therefore the steel sections W150x22 are acceptable by inspection. The additional steel will be field welded to the existing frame using 150 mm long, 6mm fillet welds at 6 locations:



As per S16-09 table 3-25, a 6mm fillet weld has a shear resistance of 0.933 kN/m. A 150 mm long weld has a shear resistance of

The total transformer weight is only 10.5 kN. Therefore, the existing steel, connections and pile foundation are acceptable by

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3.3.3 TR-417 Concrete Foundation Design

For the transformer foundation, the foundation will project 150 mm above grade.

Therefore, to protect from potential erosion / washout of the surrounding grade, use a 500 mm thick concrete foundation - 350 mm below grade (standard foundation depth used on transformers and EHT skid support in ROF).

TR-421 Foundation Dimensions

L_x	=	1500 mm	: Length of Foundation, x-dir
L_z	=	1300 mm	: Length of Foundation, z-dir
h_{FDN}	=	500 mm	: Thickness of Foundation

The foundation and soil bearing pressure was checked using RISA Foundation.

To verify the bearing pressure and concrete design, moments due to wind must be taken at the base of the foundation.

Therefore, the moments due to wind above are increased by $(V_{WT} \text{ or } V_{WL}) * 0.500 \text{ m}$:

Transverse Wind, WT: z-direction

WT [kN]	M_{WT} [kN.m]	h_{FDN} [m]	$M_{WL @ base}$ [kN.m]
1.35	1.10	0.50	1.77

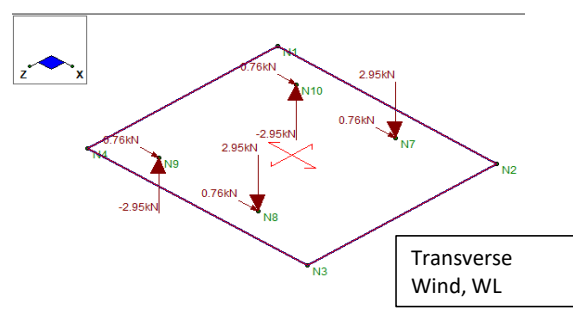
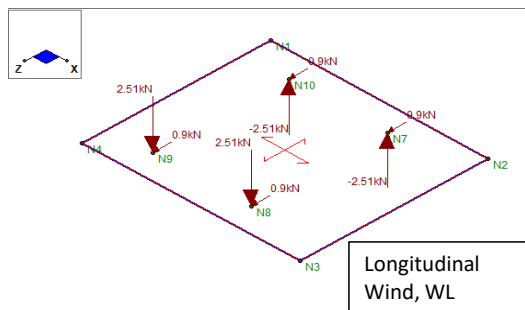
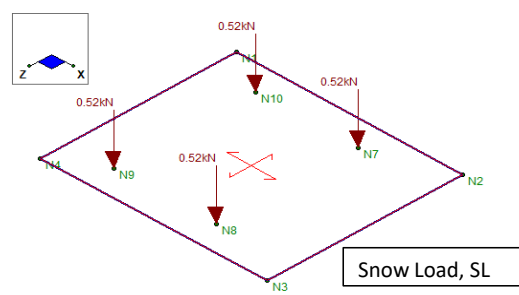
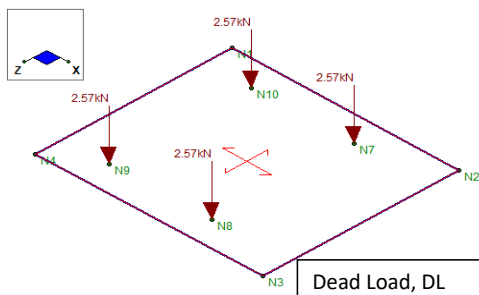
WT / Anchor [kN]	T/C* Moment [kN]	*T/C per anchor
0.34	0.94	

Longitudinal Wind, WL: x-direction

WL [kN]	M_{WL} [kN.m]	h_{FDN} [m]	$M_{WL @ base}$ [kN.m]
1.14	0.93	0.50	1.50

WL / Anchor [kN]	T/C* Moment [kN]	*T/C per anchor
0.29	1.10	

The above loading was applied in RISA Foundation:



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Reinforcement Detailing:

It is expected that the bending moment acting on the foundation is very minor.
 Therefore, use minimum reinforcement ratio:

$$A_{s,min} = 0.002A_g : A23.3-14 Cl. 7.8.1$$

$$= 1000 \text{ mm}^2/\text{m}$$

Therefore, use 15M bars @ 200 mm spacing = $A_s = 1000 \text{ mm}^2/\text{m}$, OK

FD Slab Rebar Parameters									
General Beam Mat Slab Footing/Pile Cap Pedestal/Pile									
	Label	Top Bar	Bottom Bar	Max Top Bar Sp...	Min Top Bar Sp...	Max Bot Bar Sp...	Min Bot Bar Sp...	Spacing Inc...	Rebar Options
1	Typical	#5	#5	200	200	200	200	200	Force Top and Bottom

Load Combinations:

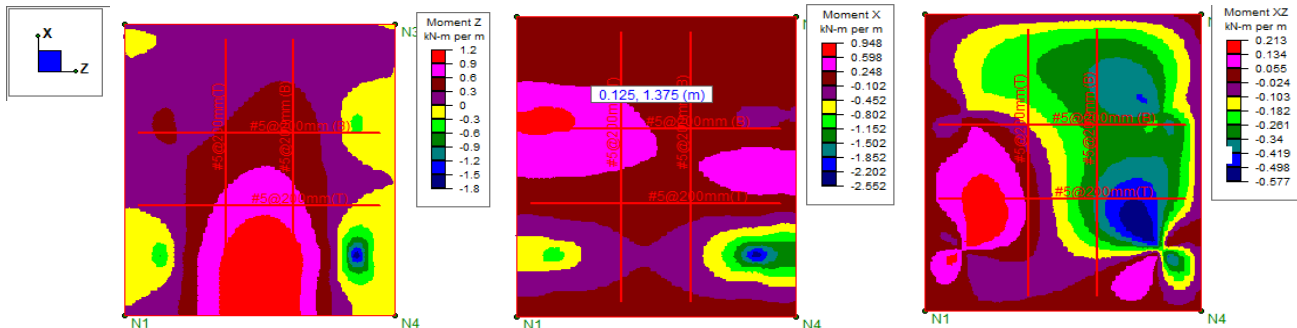
The following three load combinations were used:

Load Combinations													
	Label	Solve	Service	ABIF	SF	Categ...	Factor	Cate...	Factor	Cate...	Factor	Cate...	Factor
1	Service	☒	☒			DL	1	WLX	-1.05	WLZ	1.05		
2	Strength	☒	☐			DL	1.25	WLX	-1.05	WLZ	1.05	SL	0.5
3	Overturning	☒	☐			DL	0.9	WLX	-1.05	WLZ	1.05		

Reinforcement Check:

As can be seen, the reinforcement is acceptable for the minor bending moment experienced by the slab:

Strip Reinforcing (Envelope)											
Design Strip Results Design Cut Results											
	Label	UC Top	LC	Top Bars	Governing Design Cut f.	UC Bot	LC	Bot Bars/Mid...	Governing De...	UC Shear	LC
1	DS1	0.004	2	#5@200mm	DS1-X27	0.001	2	#5@200mm	DS1-X43	0.009	2
2	DS2	0.002	2	#5@200mm	DS2-X28	0.004	2	#5@200mm	DS2-X11	0.01	2



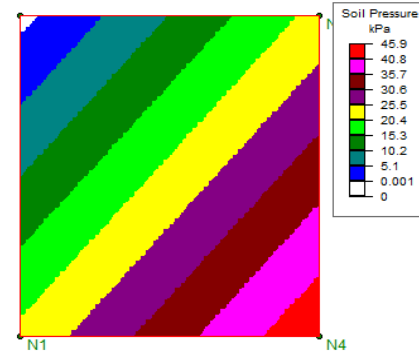
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Maximum Bearing Pressure:

As per the geotechnical report, the maximum allowable bearing pressure is limited to 150 kPa.
 The actual maximum bearing pressure is 46 kPa, OK.

Envelope Soil Pressures (By Combination)					
Soil Pressures Slab Soil Pressures Footing Soil Pressures Beam Soil Pressures					
	Label	Max UC	Max LC	Soil Pressure[kPa]	Allowable Bearing...
1	N4	0.301	2	45.122	150
2	N357	0.298	2	44.738	150
3	N2626	0.298	2	44.67	150
4	N358	0.296	2	44.354	150
5	N356	0.295	2	44.285	150
6	N2602	0.295	2	44.217	150
7	N3160	0.293	2	43.97	150
8	N359	0.293	2	43.901	150



The support slab anchor bolts, reinforcement, and bearing pressure are all acceptable.



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Appendix A - Vendor Drawings -

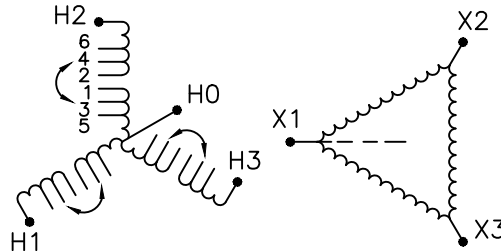
DRY TYPE POWER TRANSFORMER / TRANSFORMATEUR DE PUISSANCE DE TYPE SEC

CAT	CM6C0225 S023624	
KVA	225	
PHASE	3	60 HZ
TEMP RISE ÉLEV TEMP	115 °C	
TAPS PRISES	4 - 2.5% 2A, 2B	
ENCLOS. BOITIER	3R	1006 KG
TRANSFORMER TAG NO. TR-417 (STEP-UP)		

SECONDARY PRIMAIRE		
	VOLTAGE TENSION	CONNECTORS CONNECTEURS
	4160Y/2402	#14 to/à #2

PRIMARY SECONDAIRE		
	VOLTAGE TENSION	CONNECTORS CONNECTEURS
	480	1 of de 600 MCM

2 RUNS OF 4/0AWG



DIMENSIONS		
	IN/PO	MM
A	64.00	(1626)
B	49.00	(1245)
C	41.50	(1054)
D	30.00	(762)
E	39.00	(991)

INSULATION CLASS / CLASSE D'ISOLATION 220

* ESTIMATED VALUES / VALEURS ESTIMÉES

* **EFFICIENCY / EFFICACITÉ :

MEETS OR EXCEEDS / ÉGALE OU DÉPASSE

** - CSA : CSA C802.2-18

** - CANADA : SOR/DORS/2018-201

- QUÉBEC : O.C./DÉCRET/1394-2018

- ONTARIO : REGULATION 509/18

- USA : DOE 10 CFR PART 431:2016-01

MOUNTING : FLOOR

ANCRAGE : PLANCHER

3 PHASE / 3 COIL / COPPER

3 PHASES / 3 BOBINES / CUIVRE

IMPEDANCE / IMPÉDANCE*			please provide (135°C)				STD	%
NO LOAD LOSSES / PERTES À VIDE *							660	Watts
LOAD LOSSES / PERTES EN CHARGE * (135°C)							2178	Watts
TOTAL LOSSES / PERTES TOTALES*							2838	Watts
EXCITING CURRENT / COURANT D'EXCITATION *							1.00	%
NOISE LEVEL / NIVEAU DE BRUIT *							58	dBA
X/R RATIO / RATIO X/R *							3.33	
% RÉGULATION (135°C) *			% EFFICIENCY (135°C) *					
(FULL LOAD) (PLEINE CHARGE)			% EFFICACITÉ				(75°C)**	
1.0 P.F.	0.9 P.F.	0.8 P.F.	100%	75%	50%	25%	50%	
1.02	2.31	2.73	98.75	98.90	98.94	98.60	98.83	

REVIEWED

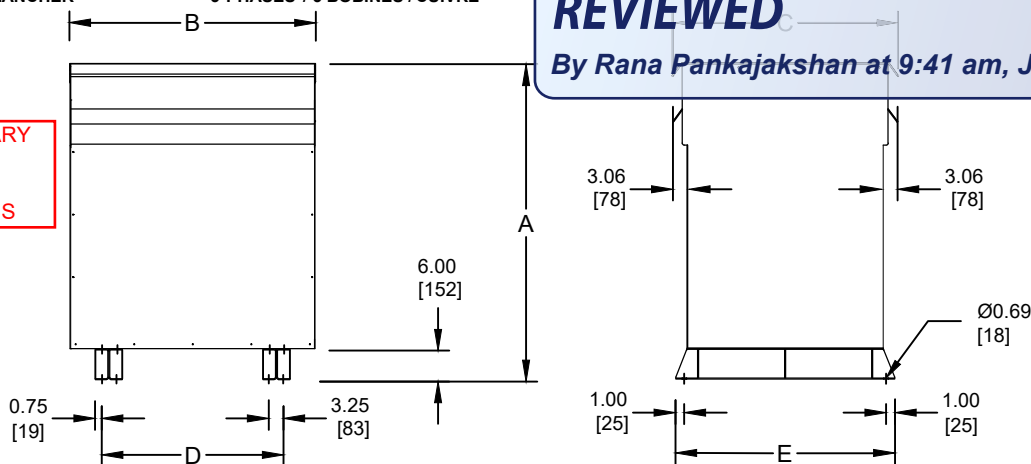
By Rana Pankajakshan at 9:41 am, Jun 16, 2021



153973



E

PLEASE INDICATE PRIMARY
AND SECONDARY
CUSTOMER CABLE
TERMINATION LOCATIONSMOUNTING : FLOOR
ANCRAGE : PLANCHER
FILE: DA - CM6C0225 S023624 R00.dwgFOR PROPER VENTILATION FOR FLOOR INSTALLATION KEEP AT LEAST 12 INCHES FROM ADJACENT WALL
POUR UNE BONNE VENTILATION LORSQUE INSTALLÉ AU SOL PLACEZ L'APPAREIL À AU MOINS 12 POUCES DES MURS ADJACENTS

CUSTOMER	Inter Pipeline Offgas Limited C/O Fluor
CLIENT	Canada
PO / BC	4000023600 / A6KV-6-1007
CONTR. PROJ.	HPC Decoupling - ROF Scope -20170

SPECIAL INST'N / INSTR. SPÉCIALE

See Page 2 for Special Instructions // Voir Page 2 pour Instructions Spéciales

WO FT	DRAWING DESSIN	CM6C0225 S023624
DRAW. DESS.	S. Roy	DATE 2021/06/11
APPD. APP.	DATE	REV. 00
		C
		T



Division Distribution	SPECIAL INSTRUCTIONS / INSTRUCTIONS SPÉCIALES CM6C0225 S023624	
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SPECIAL INSTRUCTIONS

- 30 kV BIL Insulation on SECONDARY
- Designed for Step Up Operation
- Air Filters
- 1 of 200° C N. C. Thermal Sensor in Center Coil
- Strip Heaters, 120 V Source to be Supplied by Others
- Anti-Condensation Protection (Thermo-Foil DBB) Applied Inside of Enclosure Top

INSTRUCTIONS SPÉCIALES

- Isolation 30 kV BIL au Primaire
- Conçu pour Utilisation Comme Élévateur de Tension
- Filtres à Air
- 1 Capteur Thermique 200° C N. F. Installé dans la Bobine Centrale
- Bandes Chauffantes, Source 120 V Doit Être Fournie par des Tiers
- Un Isolant Contrôlant la Condensation (Thermo-Foil DBB) Sera Installé au Toit à l'Intérieur du Boîtier

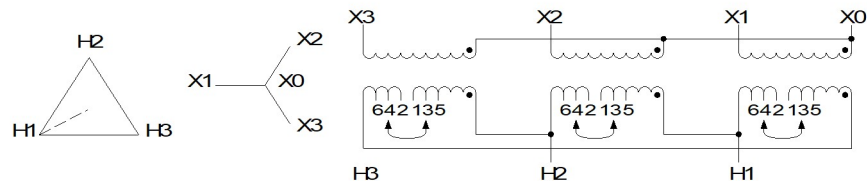
Issued by / Émis par : Sylvain Roy	Date : 2021/06/11	Page 2 of/de 3
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DRY TYPE POWER TRANSFORMER / TRANSFORMATEUR DE PUISSANCE DE TYPE SEC

CAT.	CM6C0225 S023623	
KVA	225	
PHASE	3	60 Hz
TEMP. RISE ÉLEV. TEMP.	115 °C	
TAPS PRISES	4 - 2.5 % 2A, 2B	
ENCLOS. BOITIER	3R	1047 Kg
TRANSFORMER TAG NO. TR-421 (STEP-DOWN)		

PRIMARY / PRIMAIRE		
	VOLTAGE TENSION	CONNECTORS CONNECTEURS
	4160	#14 to/à #2

SECONDARY / SECONDAIRE		
	VOLTAGE TENSION	CONNECTORS CONNECTEURS
	480Y/277	1 of/de 600 MCM

2 RUNS OF 4/0AWG

DIMENSIONS		
	IN / PO	MM
A	64	(1626)
B	49.03	(1245)
C	41.53	(1055)
D	30	(762)
E	39	(991)

INSULATION CLASS / CLASSE D'ISOLATION 220

* ESTIMATED VALUES / VALEURS ESTIMÉES

** EFFICIENCY / EFFICACITÉ :

MEETS OR EXCEEDS / ÉGALE OU DÉPASSE

- CSA : CSA C802.2-18

- CANADA : SOR/DORS/2018-201

- QUEBEC : O.C./DECRET/1394-2018

- ONTARIO : REGULATION 201/19

- USA : DOE 10 CFR PART 431:2016-01

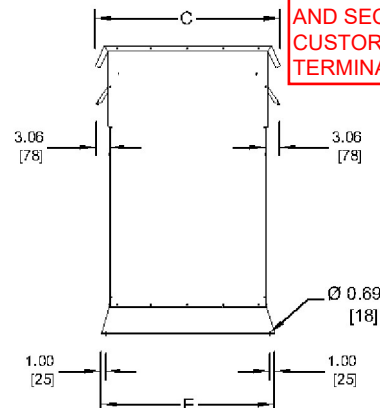
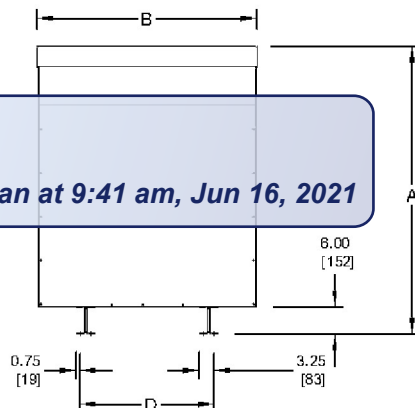
IMPEDANCE / IMPÉDANCE *					please provide		STD	(%)
NO LOAD LOSSES / PERTES À VIDE *							618	WATTS
LOAD LOSSES / PERTES EN CHARGE * (135 °C)							2167	WATTS
TOTAL LOSSES / PERTES TOTALES *							2785	WATTS
EXCITING CURRENT / COURANT D'EXCIT. *							1	(%)
NOISE LEVEL / NIVEAU DE BRUIT *							58	(dBA)
X/R RATIO / RATIO X/R *							3.85	
% REGULATION (135 °C) * (FULL LOAD) (PLEINE CHARGE)			% EFFICIENCY / % EFFICACITÉ (135 °C) * (75 °C)**					
1.0 P.F.	0.9 P.F.	0.8 P.F.	100%	75%	50%	25%	50%	
1.03	2.52	3.02	98.78	98.92	98.98	98.68	98.83	

3 PHASE / 3 COIL / COPPER

3 PHASES / 3 BOBINES / CUIVRE

REVIEWED

By Rana Pankajakshan at 9:41 am, Jun 16, 2021

**PLEASE INDICATE PRIMARY
AND SECONDARY
CUSTOMER CABLE
TERMINATION LOCATIONS**MOUNTING : FLOOR
ANCRAGE : PLANCHERFOR PROPER VENTILATION KEEP AT LEAST 12 INCHES FROM ADJACENT WALLS
POUR UNE BONNE VENTILATION PLACEZ À AU MOINS 12 POUCES DES MURS ADJACENTS

CUSTOMER CLIENT	Inter Pipeline Offgas Limited C/O Fluor Canada	SPECIAL INST'N / INSTR. SPÉCIALE	
PO / BC	4000023600 / A6KV-6-1007	See Page 2 for Special Instructions // Voir Page 2 pour Instructions Spéciales	
CONTR. PROJ.	HPC Decoupling - ROF Scope -20170		
WO FT	DRAWING DESSIN	CM6C0225 S023623 Rev. 00	
DRAW. DESS.	S. Roy	Date 2021/06/11	Ref. No 13370
APPD. APP.		Date	V0001

Tel : 1-800-663-3582
Fax : 1-877-449-9115

Division Distribution	SPECIAL INSTRUCTIONS / INSTRUCTIONS SPÉCIALES CM6C0225 S023623	
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SPECIAL INSTRUCTIONS

- 30 kV BIL Insulation on Primary
- Air Filters
- 1 of 200° C N. C. Thermal Sensor in Center Coil
- Strip Heaters, 120 V Source to be Supplied by Others
- Anti-Condensation Protection (Thermo-Foil DBB) Applied Inside of Enclosure Top

INSTRUCTIONS SPÉCIALES

- Isolation 30 kV BIL au Primaire
- Filtres à Air
- 1 Capteur Thermique 200° C N. F. Installé dans la Bobine Centrale
- Bandes Chauffantes, Source 120 V Doit Être Fournie par des Tiers
- Un Isolant Contrôlant la Condensation (Thermo-Foil DBB) Sera Installé au Toit à l'Intérieur du Boîtier

Issued by / Émis par : Sylvain Roy	Date : 2021/06/11	Page 2 of/de 3
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PROJECT: HPC Decoupling - ROF Scope Project
Location: Sturgeon County, Alberta
Title: Dry Type Transformer Foundation Design
Calculation No.: RED-00-ST-CAL-0096
By: Coltin Walsh

Contact No.: A6KV
Date: 29-Jun-2021
Check Level: 2B
Chk'd: Brett Kapcsos

Appendix B - Anchor Design -

Hilti PROFIS Engineering 3.0.70

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Company:
Address:
Phone / Fax:
Design: TR-417 - Anchorage
Fastening point:

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Specifier's comments:

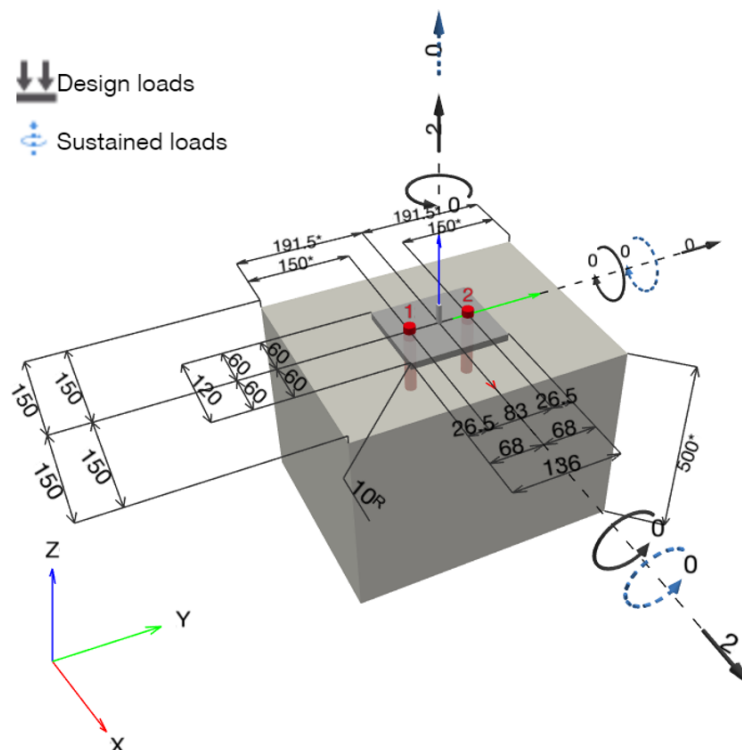
1 Input data

Anchor type and diameter:	HIT-HY 200 + HAS-V-36 (ASTM F1554 Gr.36) 5/8
Item number:	2198024 HAS-V-36 5/8"x6" (element) / 2022793 HIT-HY 200-R (adhesive)
Effective embedment depth:	$h_{ef,act} = 80.0 \text{ mm}$ ($h_{ef,limit} = - \text{mm}$)
Material:	ASTM A 1554 Grade 36
Evaluation Service Report:	ESR-3187
Issued / Valid:	- / -
Proof:	Design Method CSA A23.3-14 / Chem.
Stand-off installation:	$e_b = 0.0 \text{ mm}$ (no stand-off); $t = 10.0 \text{ mm}$
Anchor plate ^R :	$l_x \times l_y \times t = 120.0 \text{ mm} \times 136.0 \text{ mm} \times 10.0 \text{ mm}$; (Recommended plate thickness: not calculated)
Profile:	Rectangular plates and bars (AISC), 1/4 - 3/16; (L x W x T) = 6.3 mm x 4.8 mm
Base material:	cracked concrete, $f'_c = 30.00 \text{ N/mm}^2$; $h = 500.0 \text{ mm}$, Temp. short/long: 20/20 °C
Installation:	hammer drilled hole, Installation condition: Dry
Reinforcement:	tension: condition B, shear: condition B; no supplemental splitting reinforcement present edge reinforcement: none or < 15M bar



^R - The anchor calculation is based on a rigid anchor plate assumption.

Geometry [mm] & Loading [kN, kNm]



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1.1 Design results

Case	Description	Forces [kN] / Moments [kNm]	Seismic	Max. Util. Anchor [%]
1	Combination 1	N = 2.000; V _x = 2.000; V _y = 0.000; M _x = 0.000; M _y = 0.000; M _z = 0.000; N _{sus} = 0.000; M _{x,sus} = 0.000; M _{y,sus} = 0.000;	no	19

2 Load case/Resulting anchor forces

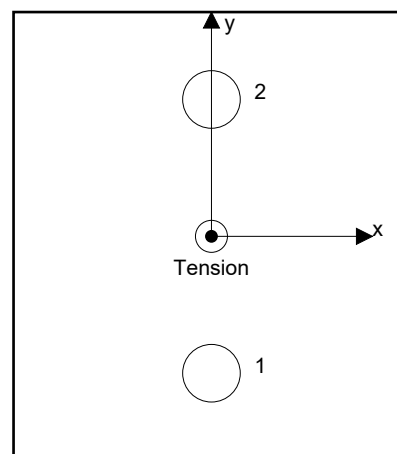
Anchor reactions [kN]

Tension force: (+Tension, -Compression)

Anchor	Tension force	Shear force	Shear force x	Shear force y
1	1.000	1.000	1.000	0.000
2	1.000	1.000	1.000	0.000

max. concrete compressive strain: - [%]
max. concrete compressive stress: - [N/mm²]
resulting tension force in (x/y)=(0.0/0.0): 2.000 [kN]
resulting compression force in (x/y)=(0.0/0.0): 0.000 [kN]

Anchor forces are calculated based on the assumption of a rigid anchor plate.



3 Tension load

	Load N _f [kN]	Capacity N _r [kN]	Utilization β _N = N _f /N _r	Status
Steel Strength*	1.000	39.655	3	OK
Bond Strength**	2.000	10.588	19	OK
Sustained Tension Load Bond Strength*	N/A	N/A	N/A	N/A
Concrete Breakout Failure**	2.000	23.999	9	OK

* highest loaded anchor **anchor group (anchors in tension)

3.1 Steel Strength

N_{s,uta} = A_{se} f_{uta} = ESR value refer to ICC-ES ESR-3187
N_{sar} = A_{se} φ_s f_{uta} R CSA A23.3-14 Eq. D.2
N_{sar} ≥ N_{fa} CSA A23.3-14 Table D.1

Variables

n	A _{se,N} [mm ²]	f _{uta} [N/mm ²]
1	146	399.90

Calculations

N _{s,uta} [kN]
58.316

Results

N _{s,uta} [kN]	φ _s	R	N _{sar} [kN]	N _{fa} [kN]
58.316	0.850	0.800	39.655	1.000

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3.2 Bond Strength

$$N_{agr} = \left(\frac{A_{Na}}{A_{Na0}} \right) \psi_{ec1,Na} \psi_{ec2,Na} \psi_{ed,Na} \psi_{cp,Na} N_{bar} \quad \text{CSA A23.3-14 Eq. D.21}$$

$$N_{agr} \geq N_{fa,g} \quad \text{CSA A23.3-14 Table D.1}$$

$$A_{Na} = \text{see CSA A23.3-14, Part D.6.5.1, Fig. D.11}$$

$$A_{Na0} = (2 c_{Na})^2 \quad \text{CSA A23.3-14 Eq. D.22}$$

$$c_{Na} = 10 d_a \sqrt{\frac{\tau_{uncr}}{7.60}} \quad \text{CSA A23.3-14 Eq. D.23}$$

$$\psi_{ec,Na} = \left(\frac{1}{1 + \frac{e_N}{c_{Na}}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.25}$$

$$\psi_{ed,Na} = 0.7 + 0.3 \left(\frac{c_{a,min}}{c_{Na}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.27}$$

$$\psi_{cp,Na} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{c_{Na}}{c_{ac}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.29}$$

$$N_{bar} = \lambda_a \phi_c \tau_r \pi d_a h_{ef} R \quad \text{CSA A23.3-14 Eq. D.24}$$

Variables

$\tau_{uncr} [\text{N/mm}^2]$	$d_a [\text{mm}]$	$h_{ef} [\text{mm}]$	$c_{a,min} [\text{mm}]$	$\alpha_{overhead}$	$\tau_{cr} [\text{N/mm}^2]$
16.18	15.9	80.0	150.0	1.000	8.53
$e'_{1,N} [\text{mm}]$	$e'_{2,N} [\text{mm}]$	$c_{ac} [\text{mm}]$	λ_a		
0.0	0.0	121.6	1.000		

Calculations

$c_{Na} [\text{mm}]$	$A_{Na} [\text{mm}^2]$	$A_{Na0} [\text{mm}^2]$	$\psi_{ed,Na}$
231.6	114,900	214,591	0.894
$\psi_{ec1,Na}$	$\psi_{ec2,Na}$	$\psi_{cp,Na}$	
1.000	1.000	1.000	

Results

ϕ_c	R	$N_{bar} [\text{kN}]$	$N_{agr} [\text{kN}]$	$N_{fa,g} [\text{kN}]$
0.650	1.000	22.113	10.588	2.000

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3.3 Concrete Breakout Failure

$$N_{cbgr} = \left(\frac{A_{Nc}}{A_{Nc0}} \right) \psi_{ec,N} \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_{br} \quad \text{CSA A23.3-14 Eq. D.4}$$

$$N_{cbgr} \geq N_{fa,g} \quad \text{CSA A23.3-14 Table D.1}$$

$$A_{Nc} \text{ see CSA A23.3-14, Part D.6.2.1, Fig. D.7}$$

$$A_{Nc0} = 9 h_{ef}^2 \quad \text{CSA A23.3-14 Eq. D.5}$$

$$\psi_{ec,N} = \left(\frac{1}{1 + \frac{2 e_N}{3 h_{ef}}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.8}$$

$$\psi_{ed,N} = 0.7 + 0.3 \left(\frac{c_{a,min}}{1.5 h_{ef}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.11}$$

$$\psi_{cp,N} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{1.5 h_{ef}}{c_{ac}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.13}$$

$$N_{br} = k_c \phi_c \lambda_a \sqrt{f_c} h_{ef}^{1.5} R \quad \text{CSA A23.3-14 Eq. D.6}$$

Variables

h_{ef} [mm]	$e'_{1,N}$ [mm]	$e'_{2,N}$ [mm]	$c_{a,min}$ [mm]	$\psi_{c,N}$
80.0	0.0	0.0	150.0	1.000
c_{ac} [mm]	k_c	λ_a	f'_c [N/mm ²]	
121.6	7.0	1.000	30.00	

Calculations

A_{Nc} [mm ²]	A_{Nc0} [mm ²]	$\psi_{ec1,N}$	$\psi_{ec2,N}$	$\psi_{ed,N}$	$\psi_{cp,N}$
77,520	57,600	1.000	1.000	1.000	1.000

Results

ϕ_c	R	N_{br} [kN]	N_{cbgr} [kN]	$N_{fa,g}$ [kN]
0.650	1.000	17.832	23.999	2.000

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4 Shear load

	Load V_f [kN]	Capacity V_r [kN]	Utilization $\beta_v = V_f/V_r$	Status
Steel Strength*	1.000	22.303	5	OK
Steel failure (with lever arm)*	N/A	N/A	N/A	N/A
Pryout Strength (Bond Strength controls)**	2.000	21.176	10	OK
Concrete edge failure in direction x+**	2.000	15.999	13	OK

* highest loaded anchor ** anchor group (relevant anchors)

4.1 Steel Strength

$V_{s,uta}$ = ESR value refer to ICC-ES ESR-3187
 $V_{sar} = 0.6 A_{se,V} \phi_s f_{uta} R$ CSA A23.3-14 Eq. D.31
 $V_{sar} \geq V_{fa}$ CSA A233-14 Table D.1

Variables

n	$A_{se,V}$ [mm ²]	f_{uta} [N/mm ²]
1	146	399.90

Calculations

$V_{s,uta}$ [kN]
34.985

Results

$V_{s,uta}$ [kN]	ϕ_s	R	V_{sar} [kN]	V_{fa} [kN]
34.985	0.850	0.750	22.303	1.000

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4.2 Pryout Strength (Bond Strength controls)

$$V_{cpgr} = k_{cp} \left[\left(\frac{A_{Na}}{A_{Na0}} \right) \psi_{ec,Na} \psi_{ed,Na} \psi_{cp,Na} N_{ba} \right] R \quad \text{CSA A23.3-14 Eq. D.45}$$

$$V_{cpgr} \geq V_{fa,g} \quad \text{CSA A23.3-14 Table D.1}$$

$$A_{Na} \text{ see CSA A23.3-14, Part D.6.5.1, Fig. D.11}$$

$$A_{Na0} = (2 c_{Na})^2 \quad \text{CSA A23.3-14 Eq. D.22}$$

$$c_{Na} = 10 d_a \sqrt{\frac{\tau_{uncr}}{7.60}} \quad \text{CSA A23.3-14 Eq. D.23}$$

$$\psi_{ec,Na} = \left(\frac{1}{1 + \frac{e_N}{c_{Na}}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.25}$$

$$\psi_{ed,Na} = 0.7 + 0.3 \left(\frac{c_{a,min}}{c_{ac}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.27}$$

$$\psi_{cp,Na} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{c_{Na}}{c_{ac}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.29}$$

$$N_{ba} = \lambda_a \phi_c \tau_r \pi d_a h_{ef} \quad \text{CSA A23.3-14 Eq. D.24}$$

Variables

k_{cp}	$\alpha_{overhead}$	$\tau_{uncr} \text{ [N/mm}^2\text{]}$	$d_a \text{ [mm]}$	$h_{ef} \text{ [mm]}$	$c_{a,min} \text{ [mm]}$
2.000	1.000	16.18	15.9	80.0	150.0
$\tau_{cr} \text{ [N/mm}^2\text{]}$	$e'_{1,N} \text{ [mm]}$	$e'_{2,N} \text{ [mm]}$	$c_{ac} \text{ [mm]}$	λ_a	
8.53	0.0	0.0	121.6	1.000	

Calculations

$c_{Na} \text{ [mm]}$	$A_{Na} \text{ [mm}^2\text{]}$	$A_{Na0} \text{ [mm}^2\text{]}$	
231.6	114,900	214,591	
$\psi_{ed,Na}$	$\psi_{ec1,Na}$	$\psi_{ec2,Na}$	$\psi_{cp,Na}$
0.894	1.000	1.000	1.000

Results

ϕ_c	$N_{ba} \text{ [kN]}$	$V_{cpgr} \text{ [kN]}$	R	$V_{cpgr} \text{ [kN]}$	$V_{fa,g} \text{ [kN]}$
0.650	22.113	21.176	1.000	21.176	2.000

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4.3 Concrete edge failure in direction x+

$$V_{cbgr} = \left(\frac{A_{Vc}}{A_{Vc0}} \right) \psi_{ec,V} \psi_{ed,V} \psi_{c,V} \psi_{h,V} \psi_{\alpha,V(D.7.2.1(c))} V_{br} \quad \text{CSA A23.3-14 Eq. D.33}$$

$$V_{cbgr} \geq V_{fa,g} \quad \text{CSA A23.3-14 Table D.1}$$

$$A_{Vc} \text{ see CSA A23.3-14, Part D.7.2.1, Fig. D.13}$$

$$A_{Vc0} = 4.5 c_{a1}^2 \quad \text{CSA A23.3-14 Eq. D.34}$$

$$\psi_{ec,V} = \left(\frac{1}{1 + \frac{2e_v}{3c_{a1}}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.38}$$

$$\psi_{ed,V} = 0.7 + 0.3 \left(\frac{c_{a2}}{1.5c_{a1}} \right) \leq 1.0 \quad \text{CSA A23.3-14 Eq. D.41}$$

$$\psi_{h,V} = \sqrt{\frac{1.5c_{a1}}{h_a}} \geq 1.0 \quad \text{CSA A23.3-14 Eq. D.42}$$

$$V_{br} = 0.58 \left(\frac{l_e}{d_a} \right)^{0.2} \sqrt{d_a} \phi_c \lambda_a \sqrt{f_c} c_{a1}^{1.5} R \quad \text{CSA A23.3-14 Eq. D.35}$$

Variables

c_{a1} [mm]	c_{a2} [mm]	e_v [mm]	$\psi_{c,V}$	h_a [mm]
150.0	150.0	0.0	1.000	500.0
l_e [mm]	λ_a	d_a [mm]	f_c [N/mm ²]	$\psi_{\alpha,V(D.7.2.1(c))}$
80.0	1.000	15.9	30.00	1.000

Calculations

A_{Vc} [mm ²]	A_{Vc0} [mm ²]	$\psi_{ec,V}$	$\psi_{ed,V}$	$\psi_{h,V}$
86,175	101,250	1.000	0.900	1.000

Results

ϕ_c	R	V_{br} [kN]	V_{cbgr} [kN]	$V_{fa,g}$ [kN]
0.650	1.000	20.887	15.999	2.000

5 Combined tension and shear loads

$\beta_N = N_f/N_r$	$\beta_V = V_f/V_r$	ζ	Utilization $\beta_{N,V}$ [%]	Status
0.189	0.125	5/3	10	OK

$$\beta_{NV} = \beta_N^{\zeta} + \beta_V^{\zeta} \leq 1$$

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6 Warnings

- The anchor design methods in PROFIS Engineering require rigid anchor plates per current regulations (AS 5216:2018, ETAG 001/Annex C, EOTA TR029 etc.). This means load re-distribution on the anchors due to elastic deformations of the anchor plate are not considered - the anchor plate is assumed to be sufficiently stiff, in order not to be deformed when subjected to the design loading. PROFIS Engineering calculates the minimum required anchor plate thickness with CBFEM to limit the stress of the anchor plate based on the assumptions explained above. The proof if the rigid anchor plate assumption is valid is not carried out by PROFIS Engineering. Input data and results must be checked for agreement with the existing conditions and for plausibility!
- Condition A applies when supplementary reinforcement is used. The R factor is increased for non-steel Design Strengths except Pullout Strength and Pryout strength. Condition B applies when supplementary reinforcement is not used and for Pullout Strength and Pryout Strength. Refer to your local standard.
- Checking the transfer of loads into the base material is required in accordance with CSA A23.3!
- Installation of Hilti adhesive anchor systems shall be performed by personnel trained to install Hilti adhesive anchors. CSA A23.3-14 Annex D, Clause D.10.1

Fastening meets the design criteria!

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7 Installation data

Profile: Rectangular plates and bars (AISC), 1/4 - 3/16; (L x W x T) = 6.3 mm x 4.8 mm

Hole diameter in the fixture: $d_f = 17.5$ mm

Plate thickness (input): 10.0 mm

Recommended plate thickness: not calculated

Drilling method: Hammer drilled

Cleaning: Compressed air cleaning of the drilled hole according to instructions for use is required

Anchor type and diameter: HIT-HY 200 + HAS-V-36

(ASTM F1554 Gr.36) 5/8

Item number: 2198024 HAS-V-36 5/8"x6" (element) / 2022793 HIT-HY 200-R (adhesive)

Maximum installation torque: 81 Nm

Hole diameter in the base material: 19.1 mm

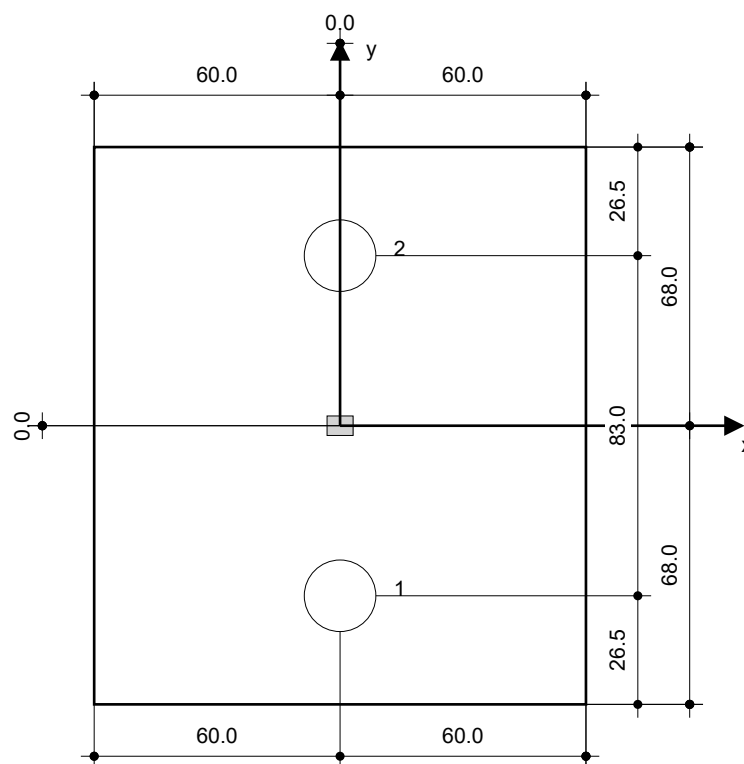
Hole depth in the base material: 80.0 mm

Minimum thickness of the base material: 118.1 mm

5/8 Hilti HAS Carbon steel threaded rod with Hilti HIT-HY 200 Safe Set System

7.1 Recommended accessories

Drilling	Cleaning	Setting
- Suitable Rotary Hammer - Properly sized drill bit	- Compressed air with required accessories to blow from the bottom of the hole - Proper diameter wire brush	- Dispenser including cassette and mixer - Torque wrench



Coordinates Anchor [mm]

Anchor	x	y	c _{-x}	c _{+x}	c _{-y}	c _{+y}
1	0.0	-41.5	150.0	150.0	150.0	233.0
2	0.0	41.5	150.0	150.0	233.0	150.0

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8 Remarks; Your Cooperation Duties

- Any and all information and data contained in the Software concern solely the use of Hilti products and are based on the principles, formulas and security regulations in accordance with Hilti's technical directions and operating, mounting and assembly instructions, etc., that must be strictly complied with by the user. All figures contained therein are average figures, and therefore use-specific tests are to be conducted prior to using the relevant Hilti product. The results of the calculations carried out by means of the Software are based essentially on the data you put in. Therefore, you bear the sole responsibility for the absence of errors, the completeness and the relevance of the data to be put in by you. Moreover, you bear sole responsibility for having the results of the calculation checked and cleared by an expert, particularly with regard to compliance with applicable norms and permits, prior to using them for your specific facility. The Software serves only as an aid to interpret norms and permits without any guarantee as to the absence of errors, the correctness and the relevance of the results or suitability for a specific application.
- You must take all necessary and reasonable steps to prevent or limit damage caused by the Software. In particular, you must arrange for the regular backup of programs and data and, if applicable, carry out the updates of the Software offered by Hilti on a regular basis. If you do not use the AutoUpdate function of the Software, you must ensure that you are using the current and thus up-to-date version of the Software in each case by carrying out manual updates via the Hilti Website. Hilti will not be liable for consequences, such as the recovery of lost or damaged data or programs, arising from a culpable breach of duty by you.